

Route and Tie a Cable Run SOP (1x Episode: 5 Clips, 3 Ties)

One episode routes one cable through all five clips, ties it at all three posts, trims all three tails, and verifies the slack loop. The table begins with the **routing board** fixed to the assembly spot, just right of the center of the table. Five open-top **clips** run left to right across the board, numbered **1** at the left to **5** at the right. An **anchor block** at the left end of the board already holds the cable's **anchored end**. Three **tie posts**, numbered **1** to **3**, stand just right of clip 1, clip 3, and clip 5. A **loop well** is printed on the board between clip 2 and clip 3, marked with a **near line** and a **far line**. A **plug rest** sits at the right end of the board. Three **pre-looped zip ties** lie in the **tie tray** at the front-right. A **flush cutter** sits in the **cutter stand** at the back-right.

The order never changes: seat the cable in all five clips, working clip 1 through clip 5; then push the slack loop forward into the loop well; then fit all three zip ties, working post 1 through post 3; then trim all three tails with the cutter and park the cutter; then verify the slack loop. No zip tie goes on until the loop is formed and every clip holds the cable. No tail is cut until all three ties are snug. The loop is verified last, with both grippers empty.

The right gripper presses the cable into every clip, forms the loop, fits and pulls every tie, cuts every tail, and does the loop check. The left gripper holds the cable — it pinches the cable just left of wherever the right gripper is working, so the cable cannot slide back, lift out of a seated clip, or drag the run. The left gripper works the left half of the board and never reaches across it. The right gripper works the board, the tie tray, and the cutter stand.

The table is set up in one of three ways. Only the loose part of the cable moves; the routing board, the anchor block, the tie tray, and the cutter stand are in the same place in all three.

- **Config L1:** the loose cable is coiled at the front-left, on the table left of the board's left end.
- **Config L2:** the loose cable is coiled at the back-left, on the table behind the anchor block.
- **Config M:** the loose cable lies along the front edge of the board in the staging zone, plug end at the front-right.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the loose cable starts coiled at the front-left (**Config L1**), coiled at the back-left (**Config L2**), or laid along the front edge of the board (**Config M**). One config per episode, chosen before recording and never changed mid-episode. The anchored end is clamped in the anchor block in all three. The right side is not used: the anchor holds the cable at the left end of the board, and the right side holds the tie tray and the cutter stand.

- **Same-side rule:** in Config L1 and L2 the **left gripper** lifts the plug end out of the coil and lays it on the board in front of clip 1, and the **right gripper** draws the cable along in front of the clips from there. In Config M each gripper takes the cable on its own half of the board. No arm reaches across the table.
- **Fixed roles:** everything else is the same in all three — the left gripper holds the cable, the right gripper seats, forms the loop, ties, trims, and checks, and the phase order above does not change.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the routing board with all five clips, all three tie posts, the anchor block, the loop well with both of its lines, the plug rest, the tie tray at the front-right, and the cutter stand at the back-right.
3. The routing board is visible from above and from the front, so the mouth of every clip, the **tie slot** under every post, the **bow** in the loop well, and every **tie head** can be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the routing board, the cable, the tie tray, the cutter stand, and robot hardware.
6. The right arm reaches clip 1, clip 5, all three tie posts, the loop well, the plug rest, the tie tray, and the cutter stand without stretching. The left arm reaches the anchor block, clip 3, and (Config L1 and L2) the coiled cable without stretching.

Materials checklist

1. One **routing board** is fixed to the **assembly spot**, just right of the center of the table, square to the front edge and flat against it. The board does not shift under a light push.
2. Five **clips** are mounted in a straight line across the board, numbered **1** at the left to **5** at the right, and the numbers are printed on the board and readable from above. Every clip mouth is **open** and points up.
3. Three **tie posts** are mounted on the board, numbered **1** to **3** left to right: post 1 just right of clip 1, post 2 just right of clip 3, post 3 just right of clip 5. Each post has an open **tie slot** under it.

4. The **loop well** is printed on the board between clip 2 and clip 3, in front of the line of clips, with a **near line** closest to the clips and a **far line** closest to the front edge of the board.
5. The **plug rest** is at the right end of the board, past clip 5, printed and open.
6. One **cable** is in place, one color, flexible, with a **plug end** at one end:
 - Its **anchored end** is already clamped in the **anchor block** at the left end of the board, and does not pull out under a light tug.
 - The rest of the cable lies loose in the **start zone** for this episode's config, not crossed over itself:
 - **Config L1**: coiled at the front-left, on the table left of the board's left end, plug end on top
 - **Config L2**: coiled at the back-left, on the table behind the anchor block, plug end on top
 - **Config M**: laid along the front edge of the board in the **staging zone**, plug end at the front-right
 - No part of the cable sits in any clip, in the loop well, or in the plug rest.
7. Three **pre-looped zip ties** lie in the **tie tray** at the front-right, heads up, each one open into a loop wide enough to pass over a tie post and the cable together.
8. One **flush cutter** sits in the **cutter stand** at the back-right, handles up, jaws down.
9. The board is clear of cable scrap and cut tail pieces.
10. Keep the left side of the table clear apart from the coiled cable in Config L1 or L2. The left gripper comes in from there onto the cable.
11. Before collection, confirm by hand that the cable presses into every clip and stays, that the cable slides along inside a clip before it is tied, that a pre-looped tie passes over a post and the cable and pulls snug, that the cutter takes a tail off without a stub, and that with the bow sitting between the two lines of the loop well the plug end still lies in the plug rest.

Workspace layout

- **Assembly spot**: just right of the center of the table — the routing board is fixed here
- **Left end of the board**: the anchor block, holding the cable's anchored end
- **Across the board**: clips 1 to 5 left to right, with tie posts 1, 2, and 3 just right of clips 1, 3, and 5
- **In front of the clips**: the loop well, between clip 2 and clip 3
- **Right end of the board**: the plug rest
- **Start zone**: where the loose cable starts — coiled at the front-left (**Config L1**), coiled at the back-left (**Config L2**), or along the front edge of the board in the staging zone (**Config M**). One per episode.
- **Front-right supply zone**: the tie tray with three pre-looped zip ties

- **Back-right tool zone:** the cutter stand
- **Left side:** kept clear apart from the coil in Config L1 or L2, so the left gripper can come in onto the cable

Arm assignments

- **Left gripper:** holds the cable. It pinches the cable just left of wherever the right gripper is working and presses it down, so the cable cannot slide back toward the anchor, lift out of a clip already seated, or drag along with the tool. It moves right along the run as the work moves right. In Config L1 and L2 it first lifts the plug end out of the coil and lays it on the board in front of clip 1.
- **Right gripper:** in Config L1 and L2, draws the cable from in front of clip 1 along the front edge of the board; in Config M, lifts the loose cable out of the staging zone. Then lays the cable into every clip, pushes the bow into the loop well, sets the plug in its rest, picks each zip tie from the tray and pulls it snug, takes the cutter and trims every tail, parks the cutter, and presses the bow for the loop check.

Vocabulary

- **Routing board:** the flat board the clips and posts are mounted on. It is fixed to the assembly spot and is never lifted or turned.
- **Clip:** one open-top holder on the board that the cable presses down into. Five of them are used in one episode.
- **Clip number:** the number printed on the board beside each clip, 1 at the left to 5 at the right.
- **Clip mouth:** the open gap at the top of a clip that the cable goes through on its way in.
- **Seated:** the cable sits all the way down at the bottom of a clip, below the clip mouth, and stays there when the right gripper lets go.
- **Cable:** the one flexible cable routed in this episode.
- **Anchor block:** the fixed clamp at the left end of the board that holds one end of the cable.
- **Anchored end:** the end of the cable held in the anchor block. It is never unclamped during an episode.
- **Plug end:** the other end of the cable, the one that is free at the start of the episode.
- **Plug rest:** the open cradle at the right end of the board that the plug end lies in at the end of the run.
- **Start zone:** where the loose cable lies at the start — coiled at the front-left (**Config L1**), coiled at the back-left (**Config L2**), or along the front edge of the board in the staging zone (**Config M**). One per episode, chosen before recording and never changed mid-episode.
- **Staging zone:** the strip along the front edge of the board where the loose cable lies at the start in Config M, and where the right gripper lays it out in Step 1 in all three configs.

- **Coil:** the loose cable wound loosely on the table with the plug end on top, in Config L1 and L2.
- **The run:** the length of cable between the anchor block and the plug rest once it is in the clips.
- **Loop well:** the marked area printed on the board between clip 2 and clip 3, in front of the clips, that the slack loop bows into.
- **Near line:** the line of the loop well closest to the clips.
- **Far line:** the line of the loop well closest to the front edge of the board.
- **Bow:** the forward curve pushed into the cable between clip 2 and clip 3.
- **Slack loop:** the bow sitting in the loop well, past the near line and not touching the far line. It is the spare cable the run needs so nothing pulls tight.
- **Springs back:** the bow returns to where it was on its own after the right gripper presses it down and lets go.
- **Tie post:** one of the three small posts on the board that a zip tie wraps around together with the cable.
- **Tie slot:** the open gap under a tie post that the zip tie loop passes through.
- **Pre-looped zip tie:** a zip tie whose tail is already started through its head, so it arrives as an open loop wide enough to drop over a post and the cable.
- **Tie head:** the small square block on a zip tie that the tail feeds through.
- **Tail:** the long flat strip of a zip tie, the part that is pulled to close the loop and is cut off later.
- **Snug:** the tie holds the cable against the post — the cable does not move along under the tie when the right gripper pushes it, and the tie does not turn around the post.
- **Clicks:** the small ticks a zip tie makes while the tail is pulled through the head. They stop when the tie will not close any further.
- **Flush cutter:** the small hand cutter used to trim tails. The right gripper closes on both of its handles to cut.
- **Cutter stand:** the holder at the back-right the cutter sits in when it is not in the right gripper.
- **Flush:** the tail is cut off at the tie head so no strip stands out past the side of the head.
- **Stub:** a short piece of tail left standing out past the tie head after a cut.
- **Kink:** a sharp bend in the cable. A kinked cable will not lie down in a clip.
- **Crossed:** the cable lies over itself somewhere along the run.

Steps

Run Steps 1–6 in order on the one cable, then end the episode with Step 7. Only Step 1 depends on the config: in Config L1 and L2 the **left gripper** lifts the plug end out of the coil and lays it in front of clip 1, and the **right gripper** draws the cable along the front edge of the board

from there; in Config M the right gripper lifts the cable out of the staging zone. Every other line is the same in all three configs.

Step 1: Take hold of the cable

Goal: the left gripper holds the cable near the anchor block, and the run is ready to lay into clip 1.

Look where the loose cable is before the first reach.

- **IF the cable is coiled at the front-left (Config L1):** with the **left gripper**, pinch the **plug end** on top of the coil, lift it straight up, carry it right onto the board, and lay it down in front of **clip 1**. Let go. With the **right gripper**, pinch the cable there and draw the plug end right along the front edge of the board until the whole loose length lies in the **staging zone** and the plug end is at the front-right. Let go.
- **IF the cable is coiled at the back-left (Config L2):** with the **left gripper**, pinch the **plug end** on top of the coil, lift it straight up, carry it forward and right onto the board, past the anchor block, and lay it down in front of **clip 1**. Let go. With the **right gripper**, pinch the cable there and draw the plug end right along the front edge of the board until the whole loose length lies in the **staging zone** and the plug end is at the front-right. Let go.
- **IF the cable lies along the front edge of the board (Config M):** with the **right gripper**, lift the loose cable out of the **staging zone** so it lies in front of the clips and does not loop back over itself.

Then, in all three:

- With the **left gripper**, pinch the **cable** just right of the **anchor block** and press it down against the board.
- Keep the **left gripper** on the cable through Steps 2, 3, 4, and 5. Move it right along the run as the work moves right, and always keep it just left of where the **right gripper** is working.

Check: the anchored end is still clamped in the anchor block, the left gripper holds the cable down, and the cable is not **crossed** over itself anywhere along its length. If the cable is crossed, lay it out straight in front of the clips with the right gripper first.

Expected state: the cable is held near the anchor, all five clip mouths are open and empty, the loop well is empty, and the plug end lies loose at the front-right.

Step 2: Seat the cable in all five clips

Goal: the cable is seated in all five clips, clip 1 through clip 5, with no bow pushed in yet.

- Work **clip 1 first, then 2, 3, 4, 5**. Do one clip at a time. Do not skip a clip and do not come back out of order.
- With the **left gripper**, hold the cable down just left of the clip being seated. After clip 1, that means holding the cable at the clip already seated.

- Do not fit any zip tie in this step. The ties stay in the tray.
- Do not push a bow into the loop well in this step.

2.1 Lay the cable into the clip

- With the **right gripper**, pinch the **cable** a little way right of the clip being seated.
- Bring the cable over the **clip mouth** and lay it straight along the line of the clips, not at an angle.
- Press the cable straight down through the clip mouth until it sits at the bottom of the clip.
- Press down only. Do not twist the cable, do not pull it along its length, and do not force a **kink** into the clip.
- Open the **right gripper** and let go. The cable must stay down in the clip.

2.2 Move along and repeat

- With the **left gripper**, move onto the cable at the clip just seated and press it down there.
- Go back to 2.1 for the next clip.

2.3 Set the plug end in its rest

- Once clip 5 holds the cable, lay the **plug end** into the **plug rest** with the **right gripper** and let go.

Check: after each clip, the cable sits below the clip mouth and stays there when the right gripper lets go. After clip 5, all five clips hold the cable, the run is not crossed over itself, and the plug end lies in the plug rest. If the cable springs back out of a clip, press it down again with the right gripper. If a clip was skipped, seat that clip before going on.

Expected state: all five clips hold the cable, the anchored end is still clamped, the plug end lies in the plug rest, the loop well is empty, and all three zip ties are still in the tray.

Step 3: Form the slack loop

Goal: a bow sits in the loop well between clip 2 and clip 3, past the near line and clear of the far line, with the plug end still in its rest.

- With the **left gripper**, hold the cable down at **clip 2**, so the slack comes from the plug side of the run and not from the anchor.
- With the **right gripper**, pinch the cable at the middle of the span between **clip 2** and **clip 3**.
- Push that span **forward, toward the front edge of the board, and down into the loop well**. The cable slides along inside clips 3, 4, and 5 as the **bow** grows.
- Keep pushing until the front of the bow sits **past the near line** and **does not touch the far line**.
- Push forward only. Do not pull the cable out of the anchor block and do not lift the cable out of a clip to make slack.

- Open the **right gripper** and let go.

Check: the bow sits between the near line and the far line, all five clips still hold the cable, and the plug end still lies in the plug rest. If the bow is short of the near line, push more cable forward. If the bow touches the far line, or the plug end has been pulled out of its rest, pinch the bow with the right gripper, draw it back toward the clips, and push it forward again to sit between the lines.

Expected state: the bow sits in the loop well between the two lines, all five clips hold the cable, the anchored end is clamped, the plug end is in its rest, and the tie tray still holds three ties.

Step 4: Fit the three zip ties

Goal: all three tie posts carry a zip tie that is snug on the cable, fitted in post order.

- Work **post 1 first, then 2, then 3**. Do one post at a time. Do not skip a post.
- With the **left gripper**, hold the cable down just left of the post being tied.
- Use one tie per post. Do not fit a second tie on a post.

4.1 Pick a zip tie

- With the **right gripper**, pinch a **pre-looped zip tie** by its **tie head** and lift it straight up out of the **tie tray**.
- Carry it in one go to the post. Do not set it down on the board on the way.

4.2 Pass the loop over the post and the cable

- With the **right gripper**, bring the open loop down over the **tie post** and the **cable** together, so the loop goes through the **tie slot** under the post.
- Let the loop settle so it sits around both the post and the cable, with the **tie head** facing up.
- If the loop catches on the post or lands around the cable only, lift it clear with the **right gripper** and bring it down again.

4.3 Pull the tie snug

- With the **right gripper**, pinch the **tail** close to the head.
- Pull the tail straight away from the post, along the board, with a slow steady pull. Listen for the **clicks**.
- Stop pulling as soon as the clicks stop. Do not keep pulling after that.
- Pull straight along the board only. Do not jerk the tail, do not pull it upward, and do not lever it against the post.
- Open the **right gripper** and let go.
- Go back to 4.1 for the next post.

Check: after each tie, the tie wraps both the post and the cable, the tie is **snug** — the cable does not move along under it when the right gripper pushes the cable, and the tie does not turn

around the post — and the bow is still sitting between the two lines of the loop well. If the tie is loose, pinch the tail again and pull until the clicks stop. If the tie went on around the cable only, cut it off with the **flush cutter**, park the cutter, and fit a fresh tie from the tray.

Expected state: all three posts carry a snug tie, three tails stick out along the board, the bow sits between the two lines, all five clips hold the cable, and the tie tray is empty.

Step 5: Trim the three tails flush

Goal: all three tails are cut off flush at their heads, with no stub left, and the cutter is back in its stand.

- Trim **post 1 first, then 2, then 3**. Cut every tail.
- Cut tails with the **flush cutter** only. Do not cut or tear a tail with the gripper tip.
- With the **left gripper**, hold the cable down just left of the tie being cut, so the cutter cannot drag the run.

5.1 Take the cutter

- With the **right gripper**, pinch the **flush cutter** by its handles and lift it straight up out of the **cutter stand**.
- Hold it jaws-down, with the flat face of the jaws toward the board.

5.2 Cut each tail

- With the **right gripper**, bring the open jaws onto the **tail** right at the **tie head**, with the flat face of the jaws against the head.
- Close the right gripper on the handles in one squeeze until the tail comes off.
- Let the cut piece drop onto the board. Do not chase it and do not pick it up.
- Look at the head. If a **stub** stands out past the side of the head, bring the jaws back onto it flat and cut again.
- Keep the jaws on the tail only. Do not let them touch the cable, a clip, or a post.
- Move on to the next tie.

5.3 Park the cutter

- Once all three tails are cut, put the **flush cutter** back in the **cutter stand** with the **right gripper**, handles up, and let go.

Check: all three heads are cut **flush**, with no stub standing out past the side of any head, the cable is not nicked or cut, all three ties are still snug, and the cutter is sitting in its stand. If a stub is left, take the cutter again and cut that head flush.

Expected state: three trimmed tie heads on the three posts, three cut pieces lying loose on the board, the cutter in its stand, and the right gripper empty.

Step 6: Verify the slack loop

Goal: the slack loop is checked once, with both grippers empty of tools, and it holds.

- With the **left gripper**, hold the cable down at **clip 2**.
- With the **right gripper**, press the top of the **bow** down toward the board with a light steady press, for a count of one.
- Open the **right gripper** and lift it clear of the loop well.
- Watch the bow. It must **spring back** and sit past the **near line** without touching the **far line**.
- If the bow stays flat, or sits short of the near line, pinch the top of the bow with the **right gripper**, lift it back up, push it forward into the loop well again, and run this check once more.
- If the loop fails this check a second time, stop, end the episode at Step 7 with the run as it is, and log the cable for a station check.

Check: the bow springs back between the two lines, no tie slid along the cable while the bow was pressed, all five clips still hold the cable, the anchored end is still clamped, and the plug end still lies in the plug rest. If a tie slid along the cable, log the run for a station check and end the episode at Step 7.

Expected state: the run sits in all five clips, three trimmed ties hold it at the three posts, and the slack loop sits in the loop well between the two lines.

Step 7: End the episode

Goal: recording ends with the cable routed, tied, trimmed, and its slack loop verified.

1. Confirm the cable is seated in all five clips, the anchored end is clamped in the anchor block, the plug end lies in the plug rest, the bow sits in the loop well past the near line and clear of the far line, all three posts carry a snug tie, all three heads are cut flush with no stub, the tie tray is empty, and the flush cutter is sitting in its stand.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. With recording off, cut all three ties off their posts with the flush cutter and drop the pieces in the scrap bin.
2. Lift the cable out of all five clips and out of the plug rest, and lay it back in the start zone for the next episode's config — coiled at the front-left (Config L1), coiled at the back-left (Config L2), or along the front edge of the board in the staging zone with the plug end at the front-right (Config M) — not crossed over itself.

3. Brush the three cut tail pieces and any tie scrap off the board and into the scrap bin.
4. Check the anchor block still holds the anchored end against a light tug. Re-clamp it if it has worked loose.
5. Lay three fresh pre-looped zip ties in the tie tray, heads up, each open wide enough to pass over a post and the cable. Put the flush cutter back in the cutter stand, handles up.
6. Inspect the parts. Replace the cable if it is nicked, cut, kinked, or has taken a set it will not straighten out of. Replace a clip that no longer holds the cable below its mouth. Replace a tie post that turns or is loose on the board. Replace the cutter if its jaws are chipped or leave a stub on a clean cut.
7. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config M (the loose cable laid along the front edge of the board, plug end at the front-right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the loose cable is not in the start zone for the config; a gripper reaches across the table for the cable; in Config L1 or L2 the right gripper takes the plug end from the coil, or the left gripper draws the cable along the board; or the wrong IF line is followed.

- **SOP rule broken:** the start position and the same-side rule (in Config L1 and L2 the left gripper lifts the plug end out of the coil and lays it in front of clip 1, and the right gripper draws the cable along the board from there; in Config M each gripper takes the cable on its own half of the board; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the loose cable is before the first reach, then follow that config's IF line through Step 1.

Violation: Cable not held

- **Visible cue:** the left gripper is off the cable, or is left behind to the right of where the right gripper is working, while the right gripper seats a clip, pushes the bow, pulls a tie, or cuts a tail — and the cable slides back, lifts out of a clip, or drags along with the tool.
- **SOP rule broken:** Step 1 (the left gripper pinches the cable and presses it down, and stays just left of where the right gripper is working through Steps 2, 3, 4, and 5).
- **Coaching note:** left gripper on the cable first, and keep it just behind the right gripper as you move right.

Violation: Routing board or fixture disturbed

- **Visible cue:** the routing board is lifted, turned, pushed off the assembly spot, or ends the episode out of square with the front edge; or the anchor block is unclamped and the anchored end comes free.
- **SOP rule broken:** Steps 1–6 (the routing board stays fixed and square on the assembly spot and is never lifted, and the anchored end stays clamped in the anchor block for the whole episode).
- **Coaching note:** bring the cable and the tool to the board. The board and the anchored end do not move.

Violation: Clips seated out of order, or a clip skipped

- **Visible cue:** the clips are seated in some order other than 1, 2, 3, 4, 5; or the episode moves on to the loop or the ties with a clip mouth still empty.
- **SOP rule broken:** Step 2 (work clip 1 first, then 2, 3, 4, 5, one clip at a time, skipping none).
- **Coaching note:** left to right, one at a time. Count five clips before you touch the loop well.

Violation: Cable not seated in a clip

- **Visible cue:** the cable sits on top of a clip mouth instead of down at the bottom of the clip, or it springs back out when the right gripper lets go and is left out.
- **SOP rule broken:** Step 2 (press the cable straight down through the clip mouth until it sits at the bottom of the clip, and let go only once it stays there).
- **Coaching note:** press it down past the mouth, let go, and look before you move on.

Violation: Cable kinked, twisted, or crossed

- **Visible cue:** the right gripper twists the cable while pressing it in, pulls it along its length, forces a kink into a clip, or lays it at an angle; or the run ends the episode lying over itself.
- **SOP rule broken:** Step 2 (lay the cable straight along the line of the clips and press down only, with no twist, no pull along its length, and no kink forced into a clip).
- **Coaching note:** lay it straight, press straight down. If it fights the clip, lift it out and lay it straight again.

Violation: Plug end left out of its rest

- **Visible cue:** the episode ends with the plug end hanging off the board, lying in the staging zone, or sitting anywhere other than the plug rest.
- **SOP rule broken:** Step 2 (lay the plug end into the plug rest once clip 5 holds the cable) and Step 3 (the plug end still lies in the plug rest once the bow is formed).
- **Coaching note:** plug goes in its rest after clip 5, and it stays there while you push the bow.

Violation: Slack loop not formed

- **Visible cue:** the run is left pulled straight between clip 2 and clip 3, with no bow pushed into the loop well, and the episode moves on to the ties or the ending.
- **SOP rule broken:** Step 3 (push the span between clip 2 and clip 3 forward and down into the loop well before any tie goes on).
- **Coaching note:** no tie goes on until there is a bow in the well. The slack is the point of the run.

Violation: Slack loop the wrong size

- **Visible cue:** the bow sits short of the near line, or it touches or crosses the far line, and the episode moves on.
- **SOP rule broken:** Step 3 (push until the front of the bow sits past the near line and does not touch the far line).
- **Coaching note:** between the two lines. Past the near line, clear of the far line.

Violation: Slack made from the wrong end

- **Visible cue:** the cable is pulled out of the anchor block, or lifted out of a seated clip, to get slack for the bow, instead of the slack coming along from the plug side.
- **SOP rule broken:** Step 3 (hold the cable at clip 2 with the left gripper and push forward only; do not pull the cable out of the anchor block or lift it out of a clip to make slack).
- **Coaching note:** hold clip 2 and push forward. The slack comes from the plug side, never from the anchor.

Violation: Ties fitted out of order, or a post skipped

- **Visible cue:** the ties go on in some order other than post 1, 2, 3; or the episode reaches the trimming or the ending with a post carrying no tie.
- **SOP rule broken:** Step 4 (work post 1 first, then 2, then 3, one post at a time, skipping none).
- **Coaching note:** left to right, all three posts. Say the post number before you pick the tie.

Violation: Zip tie fitted in the wrong place

- **Visible cue:** a tie closes around the cable only and not the post, around the post only and not the cable, around a clip, or on a post that already carries a tie.
- **SOP rule broken:** Step 4 (bring the loop down over the tie post and the cable together, through the tie slot under the post, one tie per post).
- **Coaching note:** post and cable together, through the slot. Look at what the loop is around before you pull.

Violation: Zip tie left loose

- **Visible cue:** the tail is let go before the clicks stop, or the episode moves on with a tie the cable slides under or that turns freely around its post.
- **SOP rule broken:** Step 4 (pull the tail until the clicks stop, so the cable does not move along under the tie and the tie does not turn around the post).
- **Coaching note:** pull until the clicking stops, then push the cable to check it does not move.

Violation: Tie pulled the wrong way or past snug

- **Visible cue:** the right gripper jerks the tail, pulls it upward off the board, levers it against the post, or keeps pulling after the clicks have stopped.
- **SOP rule broken:** Step 4 (pull the tail straight away from the post along the board, slow and steady, and stop as soon as the clicks stop).
- **Coaching note:** slow, straight along the board, and stop when it stops clicking.

Violation: Loop or run disturbed while tying

- **Visible cue:** while a tie is fitted or pulled, the bow is dragged out of the loop well, the cable lifts out of a clip, or the cable slides along under an earlier tie, and the episode moves on.
- **SOP rule broken:** Step 4 (hold the cable down just left of the post being tied, and keep the bow sitting between the two lines of the loop well as each tie goes on).
- **Coaching note:** hold the cable close to the post, and glance at the bow after every tie.

Violation: Tail left untrimmed

- **Visible cue:** the episode ends with one or more tails still long, sticking out along the board or off the edge.
- **SOP rule broken:** Step 5 (trim post 1, then 2, then 3; cut every tail).

- **Coaching note:** three ties, three cuts. Count the heads before you park the cutter.

Violation: Tail not cut flush

- **Visible cue:** a stub stands out past the side of a tie head after the cut, or the cut is made part way along the tail instead of at the head, and it is left that way.
- **SOP rule broken:** Step 5 (bring the jaws onto the tail right at the tie head with their flat face against the head, and cut again if a stub is left).
- **Coaching note:** jaws flat against the head, then squeeze. Look at the head before you move on.

Violation: Cutter not used, or not parked

- **Visible cue:** a tail is cut, torn, or bent off with the gripper tip instead of the cutter; or the cutter is still in the right gripper during the loop check, or is left lying on the board or the table.
- **SOP rule broken:** Step 5 (cut tails with the flush cutter only, and put it back in the cutter stand once all three tails are cut).
- **Coaching note:** cutter for tails, stand for the cutter, empty grippers for the loop check.

Violation: Cutter closed on the cable or the board

- **Visible cue:** the cutter jaws close on the cable, a clip, or a tie post, or they nick the cable while cutting a tail.
- **SOP rule broken:** Step 5 (keep the jaws on the tail only; do not let them touch the cable, a clip, or a post).
- **Coaching note:** look at what is inside the jaws before you squeeze. Tail only.

Violation: Cut piece chased

- **Visible cue:** a gripper follows a cut tail piece across the board or off the table, or sweeps at it, instead of leaving it where it fell.
- **SOP rule broken:** Step 5 (let the cut piece drop onto the board; do not chase it and do not pick it up).
- **Coaching note:** let it fall. Scrap is cleared at reset, with recording off.

Violation: Slack loop check skipped or not confirmed

- **Visible cue:** the episode goes from the trimming to the ending with the bow never pressed; or the bow is pressed and the right gripper never lifts clear to watch it spring back.
- **SOP rule broken:** Step 6 (press the top of the bow down for a count of one, lift the right gripper clear, and watch the bow spring back between the two lines).
- **Coaching note:** press, lift clear, then look. A press you do not watch is not a check.

Violation: Failed loop check not fixed and re-checked

- **Visible cue:** the bow stays flat or sits short of the near line after the press and the episode moves on; or the bow is lifted and pushed back into the well and then never checked again.
- **SOP rule broken:** Step 6 (if the bow stays flat or sits short of the near line, lift it, push it forward into the loop well again, and run the check once more).
- **Coaching note:** every re-push earns another press-and-look.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence, for example a tie goes on before all five clips hold the cable, a tie goes on before the bow is formed, a tail is cut before all three ties are snug, or the loop check is done before the tails are trimmed.
- **SOP rule broken:** Steps 2–6 (seat all five clips, form the loop, fit all three ties, trim all three tails and park the cutter, then verify the loop, in that order).
- **Coaching note:** finish the phase before you start the next one. Say the phase name before you move.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper seating a clip, pushing the bow, picking or pulling a tie, holding the cutter, or pressing the bow for the check; or the right gripper taking over the hold on the cable while the other work goes on.
- **SOP rule broken:** Steps 1–6 (the right gripper seats, forms the loop, ties, trims, and checks; the left gripper holds the cable).
- **Coaching note:** right gripper does the work, left gripper holds the cable. It does not swap.

Violation: Cable, tie, or tool knocked off its spot

- **Visible cue:** a zip tie is dropped on the board, the table, or the floor, the cutter is dropped, or the tie tray or the cutter stand is tipped, pushed off its spot, or spilled.
- **SOP rule broken:** Steps 4–5 (keep the ties, the cutter, the tie tray, and the cutter stand on their spots through the whole episode).
- **Coaching note:** work slower and lower over the table, and keep each carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the clips, the loop, the ties, and the trims are confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 7 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A zip tie's ratchet is dead out of the tray, so the tail pulls straight through the head and the tie will not hold.
- A pre-looped tie arrives closed, or its loop is too small to pass over a post and the cable.
- A clip is broken or sprung before the episode, so a correctly pressed cable will not stay below its mouth.
- A tie post is cracked or loose on the board, so a snug tie still turns.
- The cable arrives nicked, kinked, or cut, so it will not lie down in a clip or hold a bow.
- The cutter's jaws are chipped before the episode, so a clean cut still leaves a stub.

Annotation subtasks (from SOP)

1. Pinch the cable near the anchor block and hold it with the left gripper
2. Lay the cable into a clip and press it down
3. Repeat the lay and press for all five clips, clip 1 through clip 5
4. Set the plug end into the plug rest
5. Push the span between clip 2 and clip 3 forward into the loop well
6. Pick a pre-looped zip tie from the tray
7. Pass the tie loop over the post and the cable
8. Pull the tie tail snug
9. Repeat the tie fit and pull for all three posts, post 1 through post 3
10. Take the flush cutter from the stand
11. Cut each tie tail flush at its head, post 1 through post 3
12. Park the cutter back in the stand
13. Press the slack loop down and watch it spring back
14. Confirm the end state, return both arms home, and end the episode